

Is It Sorted?

Input file: **standard input**
Output file: **standard output**
Time limit: 1 second
Memory limit: 256 megabytes

Given an **array** a of n elements and a number t , your task is to **check if the array is sorted according to the value of t .**

If $t = 1$, the array must be sorted in non-decreasing order.

If $t = 2$, the array must be sorted in non-increasing order.

Input

The first line contains two integers n and t ($1 \leq n \leq 10^5$, $1 \leq t \leq 2$).

The second line contains n integers a_i ($1 \leq a_i \leq 10^9$).

Output

Print **YES** if the array is sorted according to the value of t . Otherwise, print **NO**.

Examples

standard input	standard output
5 1 1 2 5 3 4	NO
5 2 5 4 3 2 1	YES